

KARIM AOUAOU DA

Full-Stack Engineer — AI/ML — MLOps — Production Systems

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Professional Summary

Master of Artificial Intelligence candidate (graduating June 2026) with Software Engineering background and 4+ years of production-grade full-stack development. Proven expertise building secure, scalable systems with Laravel/React and desktop applications using Electron. Strong track record in CI/CD automation, cloud deployment, and architectural improvements. Transitioning AI/ML skills with hands-on PyTorch, Optuna, and MLOps workflows. Seeking international roles with technical leadership potential.

Technical Skills

Languages: PHP, JavaScript, TypeScript, Python, SQL

Frontend: React, React Native (Expo), Alpine.js, Livewire, Bootstrap, Tailwind CSS

Backend: Laravel, REST APIs, SQLCipher, MySQL

Desktop: Electron.js, CI/CD pipelines, secure app architecture

AI/ML: PyTorch, Optuna, model tuning, training automation

DevOps/Cloud: Docker, GitHub Actions, GitLab CI/CD, DigitalOcean, Hostinger

Security: Encrypted storage, session management, secure release workflows

Tools: Git, Laravel Filament, Postman, debugging

Professional Experience

Core Developer, Team Lead, Project Maintainer

Doctolik (*Medical Startup, Remote*)

Sep 2025 – Present

- Led full-stack delivery of secure medical platform with 40+ healthcare professionals; architected secure Electron desktop companion with SQLCipher encrypted storage and authenticated server synchronization
- Implemented CI/CD automation (GitLab CI) and secure release workflows; achieved zero-downtime deployments with auto-update pipeline and encrypted local data persistence
- Built payment and prescription management modules (Laravel, React); delivered 15+ critical security patches and performance optimizations
- Established technical roadmap addressing scalability, security posture, and feature backlog prioritization for cross-functional team
- Stack: [Laravel](#) | [React](#) | [Electron](#) | [SQLCipher](#) | [GitLab CI/CD](#) | [Docker](#) | [Alpine.js](#) | [Livewire](#)

Freelance Laravel Developer

Shyftcom Platform (*SaaS Startup, Remote*)

Jul 2025 – Sep 2025

- Architected and delivered core platform modules under tight startup timeline; built scalable API endpoints and database schemas supporting early product expansion
- Collaborated with founders on product structure and early-stage technical decisions
- Stack: [Laravel](#) | [MySQL](#) | [REST APIs](#) | [SQL optimization](#)

Education

Master's Degree in Artificial Intelligence

20 August 1955 University of Skikda

2024 – Jun 2026

- Advanced coursework in applied AI systems, optimization, and ML engineering
- Focus: production-ready AI systems, MLOps workflows, automated optimization pipelines

Bachelor's Degree in Software Engineering

8 May 1945 University of Guelma

2021 – Jun 2024

- Strong foundation in systems design, database architecture, and software principles

Featured Projects

Secure Desktop Companion for Doctolik

Sep 2025 – Present

- Architected cross-platform Electron application with production security standards: SQLCipher encrypted local storage, JWT authentication, and secure server synchronization for offline-first workflows
- Implemented auto-update system with cryptographic verification and signed releases; designed roll-back mechanisms for reliability
- Reduced sync latency by 40% through optimized payload compression and batch processing; ensured data integrity with ACID compliance
- Stack: [Electron.js](#) | [React](#) | [SQLCipher](#) | [CI/CD](#) | [security architecture](#)
- **Impact:** Enables 40+ medical professionals to work offline with guaranteed data consistency and security compliance

Algorithmic Trading Automation System

In Progress

- Designing robust session-based automation framework for executing trading actions on external platforms
- Implementing multi-threaded scraping logic, session persistence, and reliability mechanisms for mission-critical workflows
- Stack: [Python](#) | [Web scraping](#) | [Session management](#) | [Async programming](#)
- **Impact:** Automating high-frequency trading workflows with 99.9% uptime requirements

Additional Notable Projects

- **laravel-rater** – Multi-rater, multi-target rating system package (2 stars)
- **laravel-community** – Full community feature set: posts, comments, groups, messaging
- **laravel-quiz-platform** – Quiz platform with teacher-to-student workflows using Laravel/Livewire
- **School Manager** – Advanced school management system for private institutions and universities
- **MentorTron** – AI-powered robot mentor for university students with vision and voice interaction
- **Voice Emotion Detector** – Deep learning project built on pretrained models for emotion classification

AI/ML & MLOps Focus

- **Research Interests:** Applied AI systems, production ML workflows, optimization algorithms, intelligent automation
- **Technical Depth:** Model tuning with Optuna, PyTorch training pipelines, hyperparameter optimization, experiment tracking
- **MLOps Execution:** GitHub Actions automation, containerized inference, deployment automation, monitoring frameworks
- Actively integrating AI/ML into full-stack projects; building pipelines with training reproducibility and versioning

Why Choose Karim?

1. **End-to-end Production Ownership:** From architecture design to deployment and monitoring; proven delivery of secure, scalable systems in regulated healthcare environment
2. **Security-First Mindset:** Designed and implemented encrypted storage (SQLCIPHER), session management, secure release pipelines, and compliance workflows—critical for regulated domains
3. **CI/CD & DevOps Mastery:** Expert in GitLab CI/CD and GitHub Actions; automated testing, builds, deployments, and rollback mechanisms for zero-downtime releases
4. **AI/ML + Full-Stack Bridge:** Unique profile combining 4+ years production engineering with active AI/ML upskilling (PyTorch, Optuna, MLOps)—rare and valuable for AI-first organizations
5. **Architectural Thinking:** Designed secure Electron architecture with offline-first sync, built scalable REST APIs, optimized database schemas for performance and compliance
6. **Team Contribution:** Proven ability to lead small teams, establish technical roadmaps, and mentor on best practices while shipping features consistently

Career Direction

Primary Targets:

- **AI Engineer (Applied/Production):** Building production AI systems, optimization workflows, intelligent automation
- **MLOps Engineer:** ML infrastructure, model lifecycle, deployment automation, experiment tracking
- **Full-Stack Engineer (Laravel/React):** Technical leadership, architectural decisions, product impact

Company Types: AI-first startups, product companies, health-tech, SaaS platforms, engineering-driven teams

Preferred Roles: International (remote/onsite), with technical growth and leadership trajectory opportunities

ATS Keyword Focus

Primary: AI Engineer, MLOps Engineer, PyTorch, Laravel, React, CI/CD, Docker, Cloud Deployment, Security Architecture, Team Leadership

Secondary: Electron, SQLCIPHER, GitLab CI/CD, GitHub Actions, System Design, Web Scraping, Automation, REST APIs, Microservices, DevOps

Languages

- **Arabic:** Native fluency
- **English:** Professional working proficiency (technical writing, communication)
- **French:** Basic proficiency

Quantifiable Achievements

- **Security:** Zero security incidents in production; 100% encryption compliance for regulated health-care data
- **Performance:** 40% reduction in data sync latency; optimized API response times by 35%
- **Reliability:** Achieved 99.9% uptime across production services with zero-downtime deployment strategy
- **Development:** Reduced bug escape rate by 50% through rigorous CI/CD testing and code review practices
- **Delivery:** Shipped 15+ features and critical patches with consistent on-time delivery in fast-paced startup environment

Open Source & Community Contributions

- Published **laravel-rater** package (PHP/Laravel ecosystem) for multi-rater systems
- Active GitHub contributor with 22+ personal projects spanning full-stack, AI/ML, and DevOps domains
- Mentor experience helping university peers with system design and project architecture

Additional Information

- **Work Arrangements:** Flexible for remote, onsite, or hybrid; open to relocation for the right opportunity
- **Availability:** Immediately available (graduating June 2026; flexible with timing)
- **Interests Beyond Work:** AI systems, system design, open-source contributions, mentoring junior developers, technical writing

References and detailed portfolio available upon request.

Updated: June 8, 2026